Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Mastering Semiconductor Key Modules for Enhanced CMOS Process Integration and Device Performance

TGS Ref No: TGS-2024051248

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 8.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
8.0	20 August 2026	New Lesson Plan built from the complete v7 trainer-deck coverage; aligned to eight activities, current WSQ administration and v8.0 publication.	Dr Alfred Ang

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Course Overview

This two-day WSQ course develops the ability to identify CMOS technology process flows, analyse cross-module interactions, determine device/reliability performance, verify integration against product specifications and define follow-up action.

Field	Details
TSC	Process Integration · ELE-SIS-4002-1.1
Course code	TGS-2024051248
Duration	16 hours over 2 days
Assessment	Day 2, 4:00 PM-6:00 PM (WA 4:00-5:00 PM; PP 5:00-6:00 PM).
Trainer	Dr Alfred Ang

Learning Outcomes

- LO1: Identify process flows and modules for specific CMOS technologies in manufacturing.
- LO2: Determine CMOS device performance and verify process integration in accordance with product specifications.

Daily Schedule

Day 1 Schedule

Time	Duration	Topic / Activity	Method	Slides
9:00-9:20	20 min	Digital attendance, introductions, outcomes and assessment briefing	Facilitated orientation	Slides 1-19
9:20-10:30	70 min	Wafer fab, module interfaces and Activity 1	Visual explanation + activity	Slides 20-24
10:30-10:40	10 min	Morning break	Break	-
10:40-12:30	110 min	Photolithography, mask/resist, etch/clean and Activity 2	Concept maps + workbook	Slides 25-94
12:30-1:15	45 min	Lunch	Break	-
1:15-2:55	100 min	Ion implantation, diffusion, RTP and Activity 3	Visual explanation + analysis	Slides 95-132
2:55-3:05	10 min	Afternoon break	Break	-
3:05-5:20	135 min	Films, metallization, CMP and Activity 4	Cross-section review + activity	Slides 133-167
5:20-6:00	40 min	Integrated CMOS flows and Day 1 evidence review	Facilitated synthesis	Slides 168-207

Day 2 Schedule

Time	Duration	Topic / Activity	Method	Slides
9:00-9:15	15 min	Digital attendance and Day 1 retrieval	Recap	-
9:15-10:45	90 min	Isolation/transistor integration and Activity 5 STI review	Flow analysis + activity	Slides 208-225
10:45-10:55	10 min	Morning break	Break	-
10:55-12:30	95 min	Yield, metrology, electrical metrics and Activity 6	Worked example + workbook	Slides 226-276
12:30-1:15	45 min	Lunch	Break	-
1:15-2:45	90 min	Reliability, SPC and Activity 7 excursion response	Evidence analysis + workbook	Slides 277-338
2:45-2:55	10 min	Afternoon break	Break	-
2:55-3:40	45 min	Packaging, chiplets, FinFET/GAAFET and Activity 8	Technology gate review	Slides 339-389
3:40-4:00	20 min	Course synthesis and assessment preparation	Q&A and readiness check	Slides 390-393
4:00-5:00	60 min	Written Assessment (WA-SAQ)	Individual · open book	-
5:00-6:00	60 min	Practical Performance (PP), submission and sign-off	Individual · open book	-

Topic-by-Topic Breakdown

Topic 1 - CMOS Process Modules and Process Flow · Slides 20-207

- Wafer-fab context, cleanroom discipline and contamination control
- Photolithography, masks, resist chemistry and pattern transfer
- Wet and dry etch, selectivity, profile control and resist strip
- Ion implantation, diffusion, oxidation, anneal and rapid thermal processing
- CVD, PECVD, epitaxy, PVD, metallization and barrier/seed films
- Chemical-mechanical polishing and post-CMP cleaning
- Integrated FEOL/BEOL process flows, design rules and cross-sections

Activity 1 - Build a CMOS Process Module Map · Slide 24

Organise the legacy module coverage into a defensible FEOL-to-BEOL flow and expose the interfaces that can propagate variation.

Activity 2 - Diagnose a Pattern-Transfer Process Window · Slide 94

Connect mask/resist/exposure decisions to etch profile and select a stable corrective action without masking the root cause.

Activity 3 - Balance Implantation and Thermal Budget · Slide 132

Create an implant-to-anneal decision trail that balances activation, junction depth, damage recovery and channeling risk.

Activity 4 - Integrate Film Deposition and CMP · Slide 167

Trace deposition, fill and CMP interactions and design a verification plan that protects both planarity and electrical performance.

Topic 2 - CMOS Process Integration and Performance · Slides 208-388

- Isolation, well formation, transistor formation and interconnect integration
- Wafer sort, defect density, yield models and integration optimization
- Metrology: ellipsometry, profilometry, SEM, TEM, CD measurement and AFM
- Electrical performance: sheet resistance, gate-oxide integrity and device curves
- Reliability: hot carrier, TDDB and electromigration mechanisms and tests
- Statistical process control, capability and control-chart selection
- Packaging, heterogeneous integration, chiplets, FinFET and GAAFET

Activity 5 - Review an STI Integration Flow · Slide 225

Produce the process-flow and challenge analysis used as the foundation for Practical Performance Task 1.

Activity 6 - Analyse Yield and Process Capability · Slide 276

Recreate and improve the legacy ProcessCapability exercise using an auditable Excel workbook with dynamic formulas.

Activity 7 - Respond to an SPC CD Excursion · Slide 338

Recreate the legacy P-chart/SPC exercises and generate the corrective-action chain used in Practical Performance Task 2.

Activity 8 - Run a Future-Technology Integration Gate · Slide 389

Combine the entire course into a structured technology-gate recommendation without treating future devices as isolated marketing concepts.

Resources Required

- v8.0 PowerPoint/PDF learner slides
- v8.0 Learner Guide
- Eight self-contained activity folders with printable Markdown/PDF resources
- Pattern-transfer, capability and SPC Excel workbooks
- Projector, whiteboard/A3 paper, calculator and spreadsheet software
- LMS access for courseware and assessment submission

Assessment

Written Assessment (WA-SAQ) - 2 open-ended questions, K1-K2, 1 hour, open book. Practical Performance (PP) - 2 open-ended tasks, A1-A5, 1 hour, open book.

Day 2, 4:00 PM-6:00 PM (WA 4:00-5:00 PM; PP 5:00-6:00 PM).

Minimum 75% attendance is required. Assessment is individual and open book. Learners submit on the LMS and sign the Assessment Summary Record.

Assessment portal: https://lms-tms.tertiaryinfotech.com/